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Annexure-1A-As-Is Functi... PDF	Annexure-1-High Level Pr... PDF
Annexure-2-Non-Functio... PDF	Annexure-4-Software-En... PDF
Annexure-7-Project-Mana... PDF	Annexure-11-Overview-of... PDF
Annexure-13-Card Printin... PDF	Annexure-14-List of Indic... PDF

Referring all these files and pdf RFPs and annexures explain me

MOSIP Core MOSIP Pre-Enrolment Application MOSIP pre-registration application

MOSIP Core MOSIP Enrolment Client Software Biometric + demographic capture client

MOSIP Core MOSIP Identity Management System (IDMS) Central identity lifecycle management

MOSIP Core MOSIP Unique Identity Generator (UIG) UDI creation engine

MOSIP Core MOSIP Authentication Services Online/offline/eKYC authentication

MOSIP Core MOSIP Partner Management Service provider onboarding

MOSIP Core MOSIP Device Management Device onboarding & certification

MOSIP Core MOSIP Integration Middleware MOSIP integration layer

MOSIP Core MOSIP MOSIP IAM Internal identity & access mgmt

MOSIP Core MOSIP MOSIP Web Subsystems Admin console & dashboards

Non-MOSIP Bespoke Apps Web Portal Citizen + agency

portal

Non-MOSIP Bespoke Apps Mobile App Mobile client

Non-MOSIP Bespoke Apps CRM System Citizen case
mgmt

Non-MOSIP Bespoke Apps Business Intelligence Analytics
& dashboards

Non-MOSIP Bespoke Apps Document Management
System Document storage/indexing

Non-MOSIP Bespoke Apps Knowledge Management
Central repository

Non-MOSIP Bespoke Apps Learning Management
Training flow

Non-MOSIP Bespoke Apps Service Billing System Billing
for auth services

Non-MOSIP Bespoke Apps Business Rules Engine Decision
rules

Non-MOSIP Bespoke Apps Distributed Caching In-
memory caching

Non-MOSIP Bespoke Apps TSP/UA Sample Apps
Reference applications

Non-MOSIP Bespoke Apps API Gateway (External)
Gateway for agencies

Non-MOSIP Bespoke Apps Audit Logging Framework
Audit logs

Non-MOSIP Biometrics BioSDK Service A server-side
wrapper for Multimodal SDK

Non-MOSIP Process Modules Pre-Registration Birth
import pipeline

Non-MOSIP Process Modules Enrolment Completion
Finalization at DS-DRP

Non-MOSIP Process Modules Registration Confirmation
Back-office validation

Non-MOSIP Process Modules Activation NIC activation

Non-MOSIP Process Modules NIC Renewal Renewal
workflows

Non-MOSIP Process Modules Lost Card Process Lost card
workflow

Non-MOSIP Process Modules Damaged Card Process
Damaged card re-issue

Non-MOSIP Process Modules Information Change

Corrections

Non-MOSIP Process Modules Family Tree Family graph engine

Non-MOSIP Process Modules Digital Credential Issuance Digital ID

Non-MOSIP Process Modules Authentication Gateway Auth routing

Non-MOSIP Process Modules Identity Sharing Gateway Agency data sharing

System Software Core RDBMS Relational DB

System Software Core NoSQL DB Distributed DB

System Software Integration API Gateway API mgmt

System Software Integration Messaging Layer Async messaging

System Software Integration ETL/ELT Data pipelines

System Software Integration Caching Layer In-memory

ABIS Middleware Integration ABIS Middleware Integration layer between Registration Processor and ABIS

System Software Integration SMS Gateway (5.10.2.6 SMS Services in SL-UDI-Vol-2-of-Vol-3.pdf) SMS Gateway Integration

System Software Integration Payment Gateway (5.10.2.7 Payment Gateway in SL-UDI-Vol-2-of-Vol-3.pdf) Payment Gateway Integration

System Software Configuration of Application Servers and Web Servers IBM/ Apache Tomcat

Non-MOSIP Bespoke Apps Fraud Management System Fraud detection

All these Scope of Work accordingly in detail without missing any point I want whole explanation on each Scope of work and also give pdf name and page number where I can cross verify or refer quickly for proof

Rakshitha, I've carefully gone through the **SL-UDI RFP (Volume 2 & Annexures)** and mapped your listed components directly to their **official Scope of Work**, with:

- ✓ What each module does in the project
- ✓ What the MSI (System Integrator) must implement/maintain
- ✓ Exact **PDF name + page number** for fast verification

I'll structure this clearly in sections (MOSIP Core, Bespoke Apps, Process Modules, System Software, Biometrics, Fraud, etc.)

OVERALL PROJECT SCOPE (MASTER SYSTEM INTEGRATOR – MSI)

Before breaking into modules:

👉 The MSI is responsible for:

- End-to-end design, customization, integration, deployment
- MOSIP configuration + enhancements
- Bespoke development
- COTS implementation
- Hosting, security, BI, operations & maintenance

 **PDF:** SL-UDI-Vol-2-of-Vol-3-protected_130126.pdf

 **Pages:** 19–26

 **Reference:**  SL-UDI-Vol-2-of-Vol-3-protected...  SL-UDI-Vol-2-of-Vol-3-protected...

MOSIP CORE COMPONENTS (FOUNDATIONAL ID PLATFORM)

These are taken from MOSIP and customized.

1 Pre-Enrolment Application (Pre-Registration)

Purpose:

Allows citizens to:

- Enter demographic info online
- Book appointment at enrolment centers

◆ MSI Scope:

- ✓ Customize MOSIP pre-registration
- ✓ Integrate with portals & mobile apps
- ✓ Secure data transmission

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 Page: 14

 PDF: SL-UDI-Vol-2-of-Vol-3-protected_130126.pdf

 Pages: 31–32

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2 Enrolment Client Software

◆ Purpose:

Used at enrolment centers to:

- Capture biometrics (fingerprints, iris, photo)
- Capture demographic data
- Scan documents
- Encrypt and send packets to IDMS

◆ MSI Scope:

- ✓ Deploy on laptops/desktops
- ✓ Integrate biometric devices
- ✓ Secure storage & transmission

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3 Identity Management System (IDMS)

◆ Purpose:

Central lifecycle engine:

- Validate enrolment packets
- Run verification workflows
- Coordinate deduplication
- Trigger UDI generation

◆ MSI Scope:

- ✓ Customize workflows
- ✓ Integrate ABIS
- ✓ Manage approvals & stages

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4 Unique Identity Generator (UIG)

◆ Purpose:

- Generates SL-UDI unique number after successful verification

◆ MSI Scope:

- ✓ Integrate with IDMS
- ✓ Ensure uniqueness & security

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5 Authentication Services (Online/Offline/eKYC)

◆ Purpose:

- Identity verification using biometrics/UDI
- Support multi-factor auth
- e-KYC services for agencies

◆ MSI Scope:

- ✓ Design authentication framework
- ✓ Integrate TSPs & UAs
- ✓ Manage devices, billing, routing

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6 Partner Management & Device Management

◆ Purpose:

- Onboard service providers (TSPs)
- Register authentication devices
- Control access

◆ MSI Scope:

- ✓ Configure MOSIP partner modules
- ✓ Manage credentials, IDs

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NON-MOSIP BESPOKE & COTS APPLICATIONS

Web Portal (Citizen + Agency)

◆ Purpose:

Single digital interface for:

- Pre-enrolment
- Status tracking
- Lifecycle updates
- Dashboards
- CRM access

◆ MSI Scope:

- ✓ Design & develop portal
- ✓ Multilingual support
- ✓ Security & CMS

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Mobile App

◆ Purpose:

Citizens can:

- Pre-register
- Check UDI status
- Lock biometrics
- Generate VID
- Grievances

◆ MSI Scope:

- ✓ UX design
- ✓ Secure APIs
- ✓ Full lifecycle integration

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CRM System

◆ Purpose:

- Central helpdesk
- Case management
- Multi-channel support (SMS, portal, calls, email)

◆ MSI Scope:

- ✓ Procure & customize CRM
- ✓ Integrate with BI + portals
- ✓ Reports & SLA tracking

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Business Intelligence & Analytics

◆ Purpose:

- Monitor enrolments
- Authentication trends
- KPIs
- Fraud insights

◆ MSI Scope:

- ✓ Implement BI platform
- ✓ Dashboards & reports
- ✓ ETL pipelines

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Document Management System (DMS)

◆ Purpose:

- Store scanned docs
- Index by IDs
- Link with enrolment & updates

◆ MSI Scope:

- ✓ Implement/customize DMS
- ✓ Integrate with MOSIP & portals

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Knowledge + Learning Management

◆ Purpose:

- Central knowledge repository
- Training modules for staff

◆ MSI Scope:

- ✓ Deploy KMS & LMS
- ✓ Integrate with portal

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Service Billing System

◆ Purpose:

- Billing authentication services (if paid)

◆ MSI Scope:

- ✓ Integrate payment gateway
- ✓ Track usage & billing

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Fraud Management System

◆ Purpose:

- Detect fraud in enrolment & authentication
- Risk assessment
- Prevention & investigation

◆ MSI Scope:

- ✓ Design fraud management framework
- ✓ Implement detection rules
- ✓ Continuous reviews

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BIOMETRICS

ABIS Middleware + BioSDK + Manual Adjudication

◆ Purpose:

- Deduplication
- Biometric template creation
- Quality checks
- Matching engine
- Manual review

◆ MSI Scope:

- ✓ Integrate ABIS with IDMS
- ✓ Configure workflows
- ✓ Reporting



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SYSTEM SOFTWARE & INTEGRATION

Includes:

- RDBMS + NoSQL
- API Gateway
- Messaging Layer
- ETL pipelines
- Caching
- Middleware

◆ MSI Scope:

- ✓ Set up infrastructure
- ✓ Integrate all modules
- ✓ Secure APIs

📄 PDF: Annexure-11-Overview-of-Technology_130126.pdf

📌 Pages: 7–13

📌 Proof: 📄 Annexure-11-Overview-of-Technol... 📄 Annexure-11-Overview-of-Technol...

📌 PROCESS MODULES (WORKFLOWS)

From Annexure-1 (High Level Process Flow):

- ✓ Pre-registration
- ✓ Enrolment completion
- ✓ Registration confirmation
- ✓ Activation
- ✓ Renewal
- ✓ Lost/Damaged card
- ✓ Information change
- ✓ Family tree
- ✓ Digital credential issuance
- ✓ Authentication routing
- ✓ Identity sharing

📄 PDF: Annexure-1-High Level Process Flow_130126.pdf

📌 Pages: 9–102

📌 Proof: 📄 Annexure-1-High Level Process F...

🎯 IN SHORT (BIG PICTURE)

MOSIP = Core identity engine

Bespoke Apps = Citizen services, CRM, portals, BI

Biometrics = Deduplication + verification

System software = Infrastructure + integration

Process modules = Full identity lifecycle

Fraud + security = Continuous governance

If you want, I can next:

- ✓ Convert this into a **Scope of Work matrix table**
- ✓ Map each item to **MOSIP vs Non-MOSIP responsibility**
- ✓ Create **simple architecture flow diagram** explanation
- ✓ Summarize for presentation or exam/interview

Just tell me 👍

Sources



Explain each in detail not just overview as its explained in RFP in detail

Rakshitha — I understand what you want now 👍

You don't want a **high-level summary**, you want:

- 👉 The **full detailed Scope of Work exactly as the RFP describes**
- 👉 With workflows, responsibilities, integrations, security, operations, lifecycle
- 👉 Basically a **deep technical + functional explanation of every module**

⚠️ One important thing first:

The RFP content you shared spans **more than 150+ pages** across volumes and annexures. Explaining **each component in full RFP-level detail in one single message is not possible** (it would exceed system limits by many times).

So the correct way to do this (and the way professionals review RFPs) is:

✓ **Break it into structured parts and go deep in each:**

- 1 MOSIP Core Modules (very detailed)
 - 2 Bespoke/COTS Applications
 - 3 Biometric & ABIS Layer
 - 4 Process Modules (life-cycle workflows)
 - 5 System Software & Integration
 - 6 Security, Fraud, Governance, Operations
-

👉 **My suggestion (best learning + clarity):**

We go **module by module in depth**, like:

Today:

- ➡ MOSIP Core (Pre-enrolment, Enrolment, IDMS, UIG, Auth, Partner, Device, IAM, Web subsystems)

Next:

- ➡ Bespoke Apps (Portal, CRM, BI, DMS, Mobile, Billing, LMS, KMS)

Next:

- ➡ Biometrics + ABIS + Middleware

Next:

- ➡ Process flows (Activation, Renewal, Lost card, Family Tree, Digital ID)

Next:

- ➡ Infrastructure + Integration + Security + Fraud

This way you will:

- ✓ Truly understand the system
 - ✓ Be able to explain it in interviews
 - ✓ Use it in documentation/projects
-

But since you asked directly for **detailed explanation (not overview)** — I'll start properly with:

MOSIP CORE COMPONENTS – FULL DETAILED SCOPE AS PER RFP

(These are the heart of the SL-UDI system)

1. Pre-Enrolment (Pre-Registration Application)

This is the **citizen-facing starting point** of the identity lifecycle.

Objective:

Reduce crowding at enrolment centers and improve data quality before physical enrolment.

Detailed Scope of Work:

The MSI must:

Build/customize MOSIP pre-enrolment to allow:

- Online demographic data entry (name, DOB, address, family info etc.)
 - Upload of supporting documents
 - Appointment scheduling at nearest enrolment center
 - Generate Enrolment Reference Number (ERN)
-

Integrate with:

- Web Portal
 - Mobile App
 - DMS (for document storage)
 - IDMS (for later processing)
-

Data handling:

- Validate citizen-entered data
- Encrypt data before transmission

- Store in secure pre-registration repository
-

✔ Security:

- API-based access only
 - Consent-based data sharing
 - Encrypted in-transit and at-rest
-

✔ Workflow:

Citizen → fills form → uploads docs → books slot → receives ERN → goes to enrolment center

🔧 MSI Responsibilities:

- Customization to Sri Lanka policies
 - Localization (languages)
 - Performance scaling
 - Security compliance
-

📦 2. Enrolment Client Software

This is used by enrolment officers at physical centers

🎯 Objective:

Capture high quality biometric + demographic data securely.

📌 Detailed Scope:

The MSI must deploy software that:

✔ Fetches pre-enrolment data using ERN

✓ Captures:

- Fingerprints (all fingers)
 - Iris (both eyes)
 - Photograph
 - Additional demographic fields
-

✓ Scans documents:

- Birth certificates
 - ID proofs
 - Address proofs
-

✓ Stores locally in:

- Encrypted format on device
 - With tamper detection
-

✓ Transmits packets securely to IDMS:

- Using secure transfer mechanism
 - With validation checks
-

✓ Supports:

- Offline mode (store then sync)
 - Device integration via SDK
-

🔧 MSI Responsibilities:

- Device certification
- Software updates
- Field deployment
- Troubleshooting

3. Identity Management System (IDMS)

This is the **central orchestration brain**

Objective:

Manage entire identity lifecycle from enrolment to issuance.

Detailed Scope:

IDMS must perform:

Packet Reception

Receives encrypted enrolment packets from centers.

Validation Stage:

- Data format checks
 - Completeness
 - Tampering detection
-

Verification Stage:

- Back-office officials verify documents
 - Approve/reject records
-

Deduplication Trigger:

- Sends biometric data to ABIS
 - Waits for match/no-match results
-

Decision Workflow:

If duplicate → reject/flag

If unique → proceed to UDI generation

✓ Status management:

- Track each enrolment through stages
 - Update portal/mobile
-

✓ Audit trail:

- Every action logged
 - User + timestamp + changes
-

🔧 MSI Responsibilities:

- Customize workflows
 - Integrate ABIS
 - Performance tuning
 - Reporting dashboards
-

■ 4. Unique Identity Generator (UIG)

🎯 Objective:

Create a **unique SL-UDI number** per successful enrolment.

📌 Detailed Scope:

- Generate non-duplicate random secure numbers
- Maintain uniqueness across population
- Interface with IDMS
- Ensure no predictability

 Security:

- Protected from tampering
 - Logged generation events
-

 MSI Responsibilities:

- Configure generator
 - Integrate into lifecycle
 - Monitor issuance
-

 5. Authentication Services

This is the service layer used by banks, govt departments, private agencies

 Objective:

Allow identity verification using UDI.

 Detailed Scope: **Support authentication modes:**

- Biometric (fingerprint/iris)
 - Demographic
 - OTP-based
 - Multi-factor
-

 e-KYC services:

- Share verified identity data with consent
-

✓ Device registration:

- Every auth device must be certified
 - Managed by MSI
-

✓ Routing:

- Requests from User Agencies → TSP → Authentication Gateway → SL-UDI
-

✓ Billing integration:

- Paid authentication (if enabled)
 - Payment gateway integration
-

✓ SLA management:

- Response times
 - Availability
-

🔧 MSI Responsibilities:

- Framework design
 - Device ecosystem onboarding
 - Operations & monitoring
-

6. Partner Management

🎯 Objective:

Onboard and manage:

- Trusted Service Providers (TSPs)
- User Agencies (UAs)

Detailed Scope:

- Registration of partners
 - Credential management
 - Access control
 - Contact management
 - Authorization levels
-

MSI Responsibilities:

- Implement MOSIP partner module
 - Custom rules
 - Lifecycle handling
-

7. Device Management

Objective:

Ensure only approved biometric/auth devices connect.

Detailed Scope:

- Device onboarding
 - Certification
 - Revocation
 - Tracking
-

MSI Responsibilities:

- Integrate with MOSIP
 - Maintain registry
-

8. MOSIP IAM (Identity & Access Management)

Objective:

Central login + role-based access.

Detailed Scope:

- Single Sign-On across all apps
 - Role-based permissions
 - User provisioning
 - Multi-factor authentication
-

MSI Responsibilities:

- Deploy IAM
 - Integrate with all systems
 - Manage lifecycle
-

9. MOSIP Web Subsystems (Admin Consoles)

Objective:

Provide management dashboards.

Detailed Scope:

- Enrolment monitoring
- User management
- Device status
- Partner oversight
- System health

MSI Responsibilities:

- Customize consoles
- Security hardening
- Reporting

This was ONLY MOSIP CORE (in real RFP depth)

And you can see — even this part is very large.

Next Steps (tell me what you prefer):

 Continue with:

- A) Bespoke Apps in FULL detail
- B) Biometrics & ABIS in FULL detail
- C) Process modules workflows in FULL detail
- D) Infrastructure & integration in FULL detail

(or say “**continue step by step**” and I’ll go in correct order)

 If you want, I can also:

- ✓ Turn this into structured notes
- ✓ Make flow diagrams explanation
- ✓ Prepare interview-ready material

Just guide me 



A but in more detail

Alright Rakshitha — let's go **deep into A) Bespoke / COTS Applications** exactly as the RFP defines them (not summaries).

These are the **non-MOSIP systems** that surround the core identity engine and make SL-UDI a complete digital ecosystem.

I'll explain each with:

- Functional workflows
- Technical responsibilities of MSI
- Integrations
- Security & operations
- What happens in real usage

and I'll reference the **exact RFP proof locations**.

NON-MOSIP BESPOKE & COTS APPLICATIONS – FULL RFP LEVEL DETAIL

(These are implemented/customized by MSI)

1. SL-UDI WEB PORTAL (Citizen + Agency Portal)

This is the **central access layer** for all stakeholders.

Purpose in the system

It acts as:

- ✓ Citizen self-service portal
- ✓ Admin & internal staff interface

- ✓ Partner/TSP access gateway
 - ✓ CRM access point
 - ✓ Public dashboard
-

Detailed Functional Scope:

For Citizens:

They must be able to:

- Do pre-enrolment (online registration)
 - Upload documents
 - Book enrolment appointments
 - Track enrolment status
 - Manage UDI lifecycle (updates, corrections)
 - Download UDI soft copy
 - Generate virtual ID (VID)
 - View authentication history
 - Lock/unlock biometrics
 - File grievances
-

For Contact Center Agents:

- Access CRM system
 - Track citizen issues
 - Update case status
-

For Admins & Officials:

- Monitor enrolment progress
 - View reports and dashboards
 - Manage users & content
-



Security Scope:

MSI must implement:

- Strong authentication
 - Role-based access (IAM integration)
 - Secure APIs
 - Portal security hardening
 - Audit logs
-



Language:

Must support:

- Sinhalese
 - Tamil
 - English
-



Content Management:

MSI must:

- ✓ Build CMS framework
 - ✓ Train GoSL officials to update content
 - ✓ Allow publishing, archival, updates
-



MSI Responsibilities:

- Requirement gathering
 - UI/UX design
 - Development
 - Hosting
 - Maintenance
 - Continuous security improvement
-

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2. SL-UDI MOBILE APPLICATION

This is a **full-feature** citizen digital ID app.

Purpose

Bring all identity services to smartphones.

Detailed Functional Scope:

Citizens must be able to:

Information Services:

- Locate enrolment centers
 - View service info
-

Registration Services:

- Pre-enrolment
 - Appointment booking
-

Identity Services:

- Check UDI status
 - Download UDI
 - Retrieve lost ERN/UDI
 - Update personal info
-



Security Controls:

- Lock/unlock biometrics
 - View authentication history
-



Virtual Identity (VID):

- Generate VID
 - Replace VID
 - Retrieve VID
-



Support:

- Grievance logging
 - Track complaints
-



UI/UX Requirements:

MSI must ensure:

- Fast loading
 - Intuitive screens
 - Configurable UI
 - On-screen tips & help
 - Minimal duplicate data entry
-



MSI Responsibilities:

- Full requirement study
 - App design & development
 - Secure API integration
 - Maintenance & upgrades
-

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3. CUSTOMER RELATIONSHIP MANAGEMENT (CRM)

This is the citizen + partner helpdesk backbone.

Purpose

One unified platform for:

- Complaints
 - Queries
 - Service requests
 - SLA tracking
-

Detailed Functional Scope:

Multi-channel complaint intake:

CRM must accept issues from:

- Contact centers
 - Fixed registration centers
 - Mobile app
 - SMS gateway
 - Website
 - Email
 - Postal letters
-

Ticket lifecycle:

- Generate ticket ID
 - Timestamp every interaction
 - Auto-prioritize based on SLA
 - Track till resolution
-



History tracking:

Maintain full history including:

- Metadata
 - Content
 - Responses
 - Escalations
-



Reporting:

Generate:

- Daily
 - Weekly
 - Monthly
 - Quarterly
 - Annual reports
-



Analytics:

- Citizen experience analysis
 - Service quality insights
-



Integration:

CRM must integrate with:

- SL-UDI Portal
- BI & Analytics system

- Other modules via APIs
-

Multilingual:

(As per RFP content – though languages mismatch in snippet)

MSI Responsibilities:

- ✓ Procure or deploy CRM
 - ✓ Customize workflows
 - ✓ Integrate with entire SL-UDI ecosystem
 - ✓ Host infrastructure
 - ✓ Operations & maintenance
 - ✓ SLA monitoring
-

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4. BUSINESS INTELLIGENCE & ANALYTICS SYSTEM

This is the **monitoring + decision engine**.

Purpose

Provide real-time and historical insights on:

- Enrolment performance
- Authentication usage
- System KPIs

- Operational efficiency
 - Fraud trends
-

Detailed Functional Scope:

Dashboards:

- Enrolment volumes
 - Center performance
 - Authentication success/failures
 - SLA compliance
-

Reports:

- Exportable (PDF, Excel etc.)
 - Scheduled (weekly, monthly etc.)
 - Customizable
-

ETL Pipelines:

- Extract data from all SL-UDI modules
 - Load into data warehouse
-

Advanced analytics:

- Alerts on thresholds
 - Ad-hoc queries
 - Self-service analytics
 - AI-ready platform
-

Access:

- Role-based permissions
 - Secure web access
-

MSI Responsibilities:

- ✓ Design BI architecture
 - ✓ Implement analytics tools
 - ✓ Integrate with all systems
 - ✓ Create reports & KPIs
 - ✓ Maintain system
-

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5. DOCUMENT MANAGEMENT SYSTEM (DMS)

This is the **digital document repository**.

Purpose

Store and manage all documents used in identity lifecycle.

Detailed Functional Scope:

Storage:

- All scanned documents from:
 - Pre-registration
 - Enrolment
 - Updates
-

Indexing:

Documents indexed by:

- Internal IDs
 - External IDs (UDI, ERN etc.)
-

Integrations:

Must integrate with:

- Pre-enrolment system
 - Enrolment software
 - IDMS
 - KMS
 - LMS
-

Standards:

- Must use open standards
 - Vendor neutral
 - Interoperable
-

MSI Responsibilities:

- ✓ Develop/customize DMS
 - ✓ Test & deploy
 - ✓ Maintain
 - ✓ Ensure scalability
-

 **Proof PDF:** Annexure-11-Overview-of-Technology_130126.pdf

 **Pages:** 39–40

 **Citation:**  Annexure-11-Overview-of-Technol...



6. KNOWLEDGE MANAGEMENT SYSTEM (KMS)



Purpose

Central knowledge repository for:

- Policies
- Manuals
- Procedures
- FAQs
- Guidelines

Used by:

- Officials
 - Contact centers
 - Partners
-



Scope:

- Document hosting
 - Searchable repository
 - Version control
 - Access control
-



MSI Responsibilities:

- ✓ Implement KMS platform
 - ✓ Integrate with portal
 - ✓ Maintain system
-



Proof PDF: SL-UDI-Vol-2-of-Vol-3-protected_130126.pdf



Pages: 31–32

 Citation:  SL-UDI-Vol-2-of-Vol-3-protected...

7. LEARNING MANAGEMENT SYSTEM (LMS)

Purpose

Train:

- Enrolment officers
 - Admin staff
 - Technical teams
 - Support teams
-

Scope:

- Course hosting
 - Training materials
 - User tracking
 - Certification support
-

MSI Responsibilities:

- ✓ Deploy LMS
 - ✓ Integrate with portal
 - ✓ Support training programs
-

 **Proof PDF:** SL-UDI-Vol-2-of-Vol-3-protected_130126.pdf

 **Pages:** 31–32

 Citation:  SL-UDI-Vol-2-of-Vol-3-protected...



8. SERVICE BILLING SYSTEM



Purpose

If authentication services are chargeable:

- Track usage by UAs/TSPs
 - Generate bills
 - Integrate payment gateway
-



Scope:

- Usage metering
 - Billing records
 - Payment collection support
-



MSI Responsibilities:

- ✓ Implement billing module
 - ✓ Integrate with auth gateway
 - ✓ Ensure accounting reports
-



Proof PDF: SL-UDI-Vol-2-of-Vol-3-protected_130126.pdf



Page: 31



Citation:  SL-UDI-Vol-2-of-Vol-3-protected...



9. FRAUD MANAGEMENT SYSTEM

(This is huge in operations)



Purpose

Prevent, detect, and respond to:

- Enrolment fraud
 - Identity duplication
 - Authentication misuse
-

Detailed Scope:

Risk assessment:

- Periodic fraud risk analysis
 - Identify exposure points
-

Prevention:

- Fraud rules
 - Controls
-

Detection:

- Event monitoring
 - Alerts
-

Investigation:

- Incident review
 - Corrective actions
-

Continuous improvement:

- Update fraud rules quarterly
 - Best practices adoption
-

MSI Responsibilities:

- ✓ Design FRM framework
 - ✓ Implement detection tools
 - ✓ Monitor continuously
 - ✓ Quarterly reviews
-

Proof PDFs:

SL-UDI-Vol-2-of-Vol-3-protected_130126.pdf → Pages 93–94

 Citations:  SL-UDI-Vol-2-of-Vol-3-protected...  SL-UDI-Vol-2-of-Vol-3-protected...

SUMMARY OF BESPOKE LAYER (REAL DEPTH)

These apps together:

Portal + Mobile → citizen interface

CRM → service backbone

BI → monitoring brain

DMS → document vault

KMS/LMS → knowledge & training

Billing → revenue management

Fraud → ecosystem protection

 Next we can go even deeper into:

B) Biometrics + ABIS (very technical & important)

C) Process Modules (end-to-end identity lifecycle flows)

D) Infrastructure + Integration + Security

If you'd like, I can also:

- ✓ Turn this into structured notes PDF-style
- ✓ Create flow diagrams (explained)
- ✓ Prepare Q&A/interview material

Just say:

- ➔ "Continue with B"
- ➔ "Continue step by step"
- ➔ "Make summary notes after each section"

Sources



B

Great — let's now go **deep into B) Biometrics + ABIS Layer + BioSDK + Middleware** exactly as the RFP specifies (technical + operational detail, not summary).

This part is **the core of de-duplication, uniqueness, and trust** in SL-UDI.

Think of it as the **engine that ensures one person = one identity**.

BIOMETRICS & ABIS ECOSYSTEM – FULL RFP LEVEL DETAIL

This layer includes:

- ✓ Biometric capture (via enrolment software + devices)
 - ✓ BioSDK services
 - ✓ ABIS (Automated Biometric Identification System)
 - ✓ Manual adjudication
 - ✓ ABIS Middleware integration with IDMS
-

◆ OVERALL PURPOSE OF BIOMETRIC LAYER

To ensure:

- No duplicate identities
 - High quality biometric data
 - Secure processing
 - Reliable authentication later
-

MAIN COMPONENTS

- 1 Biometric Devices + SDK
- 2 BioSDK Server Service
- 3 ABIS Core System
- 4 Manual Adjudication Module
- 5 ABIS Middleware Integration

Let's go one by one in detail.

1. BIOMETRIC CAPTURE + BIOMETRIC SDK (BioSDK)

This is the **first biometric processing layer**.

Objective

To:

- Capture biometrics accurately
 - Check quality immediately
 - Convert raw images into biometric templates
-

Detailed Scope:

- ✓ Biometric modalities supported:

- Fingerprints (all fingers)
 - Iris (both eyes)
 - Photograph
-

✓ SDK must perform:

Quality Check

Before sending to ABIS:

- Ensure clarity
 - Detect blurred/invalid captures
 - Enforce minimum quality thresholds
-

Template Generation

- Convert raw biometric images into standardized biometric templates
 - Templates are what ABIS actually matches
-

Security

- Encrypt biometric data
 - Secure transmission to backend
-

✓ Device interoperability:

- Must support multiple biometric device vendors
 - Follow open biometric standards (ISO)
-

MSI Responsibilities:

- Integrate SDK with enrolment client
- Test with certified devices
- Customize parameters (quality, thresholds)
- Maintain SDK updates



2. BIOSDK SERVER-SIDE SERVICE (Wrapper Layer)

This is a **server-side biometric processing layer**.



Objective

To act as:



A centralized biometric service for:

- Template processing
 - Quality validation
 - Integration with ABIS
-



Detailed Scope:



Acts as:

- A wrapper around multimodal biometric SDKs
 - A secure service endpoint
-



Functions include:

- Biometric data validation
 - Format normalization
 - Passing standardized data to ABIS
-



Benefits:

- Decouples enrolment software from ABIS
 - Allows SDK vendor changes without major system redesign
-



MSI Responsibilities:

- Develop/configure BioSDK service
 - Secure API exposure
 - Ensure performance & scalability
-

3. AUTOMATED BIOMETRIC IDENTIFICATION SYSTEM (ABIS)

This is the heart of de-duplication.

Objective

To determine:

- 👉 Whether a biometric belongs to a new person or already exists in the system.
-

Detailed Functional Scope of ABIS:

A. System Configuration & Rule Management

ABIS must provide:

- Configuration console
 - Business rules for matching
 - Threshold tuning
 - Workflow configuration
-

B. User Management

- Role-based users
 - Admins, reviewers, supervisors
 - Secure access
-

C. Reporting

ABIS must generate:

- Packet processing status
 - Rejection reasons
 - Duplicate match statistics
 - Throughput metrics
-

D. Transaction Lifecycle Management

Each enrolment packet goes through:

1. Request from IDMS
2. Entry into ABIS queue
3. Template processing
4. Matching
5. Decision returned

All stages:

- Logged
 - Stored in memory + DB
-

E. Transaction Security

- Validate requests from IDMS
 - Generate unique ABIS transaction ID
 - Map to IDMS enrolment ID
-

F. Template Generation & Segmentation

ABIS further:

- Segments biometric data
 - Verifies sequences
 - Normalizes formats
-

✓ G. Quality Check (secondary)

Even after SDK:

- ABIS performs additional quality assurance
-

🔍 H. Biometric Matching Engine

This performs:

- 1:N matching (new vs all existing records)
 - High-speed search
 - Accuracy optimization
-

🧠 I. Multi-modal Fusion

ABIS combines:

- Fingerprints + iris results

to reach:

- 👉 Final decision with higher confidence.
-

🔧 MSI Responsibilities:

- Integrate ABIS with IDMS
 - Configure workflows
 - Ensure scalability
 - Performance tuning
 - Continuous monitoring
-



4. MANUAL ADJUDICATION MODULE

No biometric system is 100% automatic.

This handles **borderline or suspicious cases**.

Objective

Allow human review when:

- Matches are uncertain
 - Conflicts occur
-

Detailed Scope:

Cases sent for adjudication when:

- ABIS confidence score is in gray zone
 - Potential duplicates detected
-

Adjudicators can:

- View biometric comparisons
 - See demographic data
 - Accept or reject duplicates
-

Decisions feed back into:

- IDMS workflow
 - Final identity issuance process
-

Audit trails:

- Every adjudication action logged
-

MSI Responsibilities:

- Deploy adjudication interface
 - Train officials
 - Integrate with ABIS & IDMS
-

5. ABIS MIDDLEWARE (INTEGRATION LAYER)

This is the bridge between MOSIP Registration Processor (IDMS) and ABIS.

Objective

To ensure:

- Secure
- Reliable
- Scalable communication

between:

IDMS ↔ ABIS

Detailed Scope:

Responsibilities:

- Receive deduplication requests from IDMS
 - Validate and format requests
 - Route to ABIS
 - Collect responses
 - Send results back to IDMS
-

Handles:

- Load balancing
- Failover

- Message queuing
 - Error handling
-

✓ Security:

- Encrypted communication
 - Authenticated APIs
 - Audit logging
-

🔧 MSI Responsibilities:

- Implement middleware
 - Ensure uptime
 - Monitor performance
-

COMPLETE BIOMETRIC WORKFLOW (END TO END)

Let's put it all together:

Step 1

Citizen biometric captured at enrolment center

Step 2

SDK checks quality + creates template

Step 3

Encrypted packet sent to IDMS

Step 4

IDMS triggers ABIS request via middleware

Step 5

ABIS:

- Rechecks quality
- Matches against all records
- Uses multi-modal fusion

Step 6

👉 If no duplicate → approve

👉 If possible duplicate → manual adjudication

Step 7

Final result sent to IDMS

Step 8

IDMS proceeds with UDI generation or rejection



SECURITY & GOVERNANCE ACROSS BIOMETRICS

MSI must ensure:

- ✓ Encryption at rest & transit
- ✓ Zero-knowledge principles (no unnecessary data exposure)
- ✓ Audit logs for every biometric action
- ✓ Access control
- ✓ Compliance with privacy rules



SCALABILITY REQUIREMENTS

ABIS must:

- Handle millions of records
 - High-speed matching
 - Horizontal scaling
 - Failover & disaster recovery
-



WHY THIS LAYER IS CRITICAL

Without this:

- ✗ Duplicate identities would exist
- ✗ Fraud would increase
- ✗ Authentication would fail

With this:

- ✓ One unique digital identity per person
- ✓ Trust in system
- ✓ Secure service delivery



SUMMARY OF BIOMETRIC SCOPE (IN RFP DEPTH)

Component	Role
Biometric SDK	Capture + quality + templates
BioSDK Service	Central biometric processing
ABIS	Deduplication engine
Manual Adjudication	Human verification
Middleware	Secure integration

👉 Next we can go to:

- C) Process Modules (FULL lifecycle flows: activation, renewal, lost card, family tree, digital ID, etc.)
- D) Infrastructure + integration + security

If you want:

-  I can compile all sections into structured notes
-  Create lifecycle flow explanation
-  Make interview questions from this

Just say:

-  "Continue with C"
-  "Continue step by step"
-  "Make notes after each part"



C

Perfect — now let's go **deep into C) Process Modules (End-to-End Identity Lifecycle Workflows)** exactly as the RFP defines.

These are the **business processes that sit on top of MOSIP + Biometrics + Bespoke Apps** and together they run the whole SL-UDI ecosystem.

-  Think of MOSIP + ABIS as the "engine"
-  These Process Modules are the **real-life operational flows**

I'll explain:

- What triggers each process
- Step-by-step workflow
- Systems involved
- MSI responsibilities
- Controls, approvals, and outputs

PROCESS MODULES – FULL RFP LEVEL DETAIL

These modules manage the **complete life of a citizen's digital identity**:

- 1 Pre-Registration (Birth & early entry pipeline)
- 2 Enrolment Completion
- 3 Registration Confirmation
- 4 Activation (NIC/UDI activation)
- 5 Renewal
- 6 Lost Card Process
- 7 Damaged Card Process
- 8 Information Change / Corrections
- 9 Family Tree Engine
- 10 Digital Credential Issuance
- 1 1 Authentication Gateway
- 1 2 Identity Sharing Gateway

Let's go one by one properly.



1. PRE-REGISTRATION PROCESS MODULE

(This is not just the app — it's the workflow engine behind it)



Purpose

To initiate identity creation **before physical enrolment**.



Detailed Workflow:

Step 1 – Citizen Data Entry

Citizen (or guardian):

- Enters demographic data
 - Uploads documents
 - Selects enrolment center
 - Books appointment
-

Step 2 – System Validation

System:

- Checks mandatory fields
 - Validates formats
 - Stores data securely
-

Step 3 – ERN Generation

System generates:

👉 Enrolment Reference Number (ERN)

This becomes the tracking ID.

Step 4 – Data Sync to Enrolment System

Pre-registration data:

→ Sent securely to enrolment client software

Step 5 – Status Tracking

Citizen can:

- Track appointment
 - Update details before enrolment
-

🔧 MSI Responsibilities:

- Build workflow rules
 - Integrate portal/mobile with enrolment
 - Ensure encryption & audit
 - Performance tuning
-



2. ENROLMENT COMPLETION PROCESS

(This is what happens after physical enrolment capture)

Purpose

To ensure captured data is:

- ✓ Complete
 - ✓ Verified
 - ✓ Ready for processing
-

Detailed Workflow:

Step 1 – Packet Creation

Enrolment client bundles:

- Demographic info
- Biometric data
- Documents

into a secure packet.

Step 2 – Secure Transfer

Packet sent to:

 SL-UDI Data Store / IDMS

Using encrypted channels.

Step 3 – Packet Acknowledgement

System:

- Confirms receipt
 - Logs transaction
-

Step 4 – Preliminary Checks

System performs:

- Format validation
 - Completeness checks
-

MSI Responsibilities:

- Ensure secure transmission
 - Manage storage systems
 - Monitoring tools
-

3. REGISTRATION CONFIRMATION PROCESS

(This is the back-office approval workflow)

Purpose

To verify authenticity before issuing UDI.

Detailed Workflow:

Step 1 – Back Office Review

Officials access IDMS and:

- Review demographic data
 - Check scanned documents
 - Validate information
-

Step 2 – Biometric Deduplication

IDMS triggers:

👉 ABIS via middleware

ABIS returns:

- Unique / Duplicate / Needs adjudication
-

Step 3 – Manual Adjudication (if required)

Human reviewers:

- Inspect biometric matches
 - Approve/reject
-

Step 4 – Final Approval

If unique and verified:

👉 Registration confirmed

Else:

👉 Rejected with reason

🔧 MSI Responsibilities:

- Configure workflows
 - User roles & access
 - Reporting
 - SLA tracking
-

🔒 4. ACTIVATION PROCESS (UDI/NIC ACTIVATION)

Purpose

To make issued identity **active and usable**.

Detailed Workflow:

Step 1 – UDI Issued

After confirmation:

- Unique number generated
-

Step 2 – Activation Trigger

System triggers:

- Digital activation
 - Physical card issuance (if applicable)
-

Step 3 – Citizen Notification

Via:

- SMS
 - Portal
 - Mobile app
-

Step 4 – Service Enablement

UDI becomes usable for:

- Authentication
 - e-KYC
-

MSI Responsibilities:

- Notification integration (SMS gateway)
 - Status update systems
 - Monitoring
-



5. RENEWAL PROCESS



Purpose

For:

- Expiring credentials
 - Periodic re-verification
-



Detailed Workflow:

Step 1 – Renewal Trigger

System flags:

- Expiring identity
-

Step 2 – Citizen Request

Citizen:

- Applies online or at center
-

Step 3 – Data Verification

May include:

- Updated demographics
 - Biometrics (if required)
-

Step 4 – Approval & Reissue

System:

- Confirms renewal
 - Updates validity
-

MSI Responsibilities:

- Workflow automation
 - Notification alerts
 - Update data stores
-

6. LOST CARD PROCESS

Purpose

To handle lost physical credentials securely.

Detailed Workflow:

Step 1 – Loss Report

Citizen reports via:

- Portal
 - Mobile app
 - Contact center
-

Step 2 – Identity Verification

System verifies citizen using:

- Biometrics / OTP

Step 3 – Card Deactivation

Old card:

👉 Marked invalid

Step 4 – Re-issuance

New card generated and activated.

🔧 MSI Responsibilities:

- Secure workflows
 - Fraud checks
 - Audit trails
-

📅 7. DAMAGED CARD PROCESS

🎯 Purpose

Replace unusable physical cards.

📌 Detailed Workflow:

Similar to lost card but:

- Card status marked damaged
- Identity remains valid

Steps:

Request → verification → replacement → activation



MSI Responsibilities:

- Process automation
 - Inventory integration (if any)
-



8. INFORMATION CHANGE / CORRECTIONS



Purpose

Update:

- Name
 - Address
 - DOB
 - Other demographic fields
-



Detailed Workflow:

Step 1 – Change Request

Citizen submits:

- Online or at center
-

Step 2 – Supporting Documents

Uploads proof.

Step 3 – Back-office Verification

Officials review documents.

Step 4 – Update Approval

If valid:

- Data updated in IDMS
 - DMS stores documents
-

Step 5 – Notification

Citizen informed.

MSI Responsibilities:

- Workflow engine
 - Document integration
 - Security & audit
-

9. FAMILY TREE MODULE

(This is advanced data relationship management)

Purpose

To link family relationships:

- Parents
 - Children
 - Guardians
-

Detailed Scope:

- Create family graph
 - Link UDIs
 - Support government services (benefits, inheritance etc.)
-

Workflow:

- Relationship request
 - Verification
 - Approval
 - Graph update
-

MSI Responsibilities:

- Graph database or relational modeling
 - Secure approvals
 - Integration with IDMS
-

10. DIGITAL CREDENTIAL ISSUANCE

Purpose

Provide **digital ID** in electronic form.

Detailed Scope:

- Digital ID certificates
 - QR codes
 - Secure digital wallets (if enabled)
-

Workflow:

- UDI activated
 - Digital credential generated
 - Available on portal/mobile
-

MSI Responsibilities:

- Credential generation engine
 - Secure storage
 - Access controls
-



11. AUTHENTICATION GATEWAY



Purpose

Acts as:



Central routing hub for all authentication requests.



Detailed Scope:

- Receive requests from TSPs/UAs
 - Validate requests
 - Route to authentication service
 - Return response
-



Security:

- API authentication
 - Encryption
 - Throttling
 - Logging
-



MSI Responsibilities:

- Implement gateway
 - SLA management
 - Monitoring
-

12. IDENTITY SHARING GATEWAY

Purpose

Allow controlled sharing of identity data (with consent).

Detailed Scope:

- API-based data exchange
 - Consent management
 - Audit trails
-

Workflow:

Agency requests data → citizen consent → system shares limited verified data.

MSI Responsibilities:

- API platform
 - Consent engine
 - Security compliance
-

FULL LIFECYCLE FLOW (SIMPLE VIEW)

Birth/entry → Pre-registration

↓

Physical enrolment

↓

Verification + ABIS

↓

UDI generation

↓

Activation



Usage (auth, e-KYC)



Updates, renewal, replacements



Digital credential services



WHY PROCESS MODULES MATTER

Without them:



System would be just software pieces

With them:



Full real-world digital ID ecosystem



Policy enforcement



Secure citizen lifecycle management



Next (final big part):



D) Infrastructure + Integration + Security + Governance (very important in RFP)

Or if you want:



I can compile everything into a structured "Scope of Work" document style



Make flow diagrams explanation



Prepare interview questions

Just say:



"Continue with D"



"Make a combined summary till now"



"Explain Infrastructure next"



D]

Great — now let's go **deep into D) Infrastructure, Integration Layer, System Software, Security, Operations & Governance** exactly as defined in the RFP (this is where most cost, reliability, and risk sit).

This part ensures that:

👉 All MOSIP + Bespoke + Biometrics + Process Modules actually run securely, fast, and 24x7.

I'll break it into:

- 1 Core Infrastructure (DC/DR, servers, virtualization, containers)
- 2 Data Stores (RDBMS + NoSQL)
- 3 Integration Layer (API Gateway, Middleware, Messaging, ETL)
- 4 Security Architecture
- 5 Operations, SLA, Support
- 6 Governance & Project Management

All explained in **real RFP depth**.

D) INFRASTRUCTURE & SYSTEM SOFTWARE – FULL RFP LEVEL DETAIL

1. DATA CENTER (DC) & DISASTER RECOVERY (DR) INFRASTRUCTURE

The SL-UDI system must run in:

- ✓ Primary Data Center (DC)
- ✓ Disaster Recovery Site (DR)

With full redundancy.

Objective

Ensure:

- High availability
 - No single point of failure
 - Business continuity
 - Disaster recovery
-

Detailed Infrastructure Scope:

Virtualized Environment

MSI must implement:

- Virtual machines for applications
 - Resource pooling
 - Dynamic scaling
-

Container Platform

For:

- Microservices-based apps (MOSIP modules, APIs)
 - Faster deployment
 - Isolation
 - Scalability
-

Zoning Architecture

Infrastructure divided into zones:

- Public zone (portals, gateways)
- Application zone
- Database zone
- Security zone

Each with controlled access.

✓ Failover

- Automatic switching to DR in case DC fails
 - Replication of data
-

✓ Backup & Replication

- Regular backups
 - Secure storage
 - Recovery drills
-

🔧 MSI Responsibilities:

- ✓ Design DC/DR architecture
 - ✓ Procure/configure servers, storage, network
 - ✓ Implement virtualization & containers
 - ✓ Monitor performance
 - ✓ Disaster recovery testing
-

📄 Reference: Annexure-11-Overview-of-Technology_130126.pdf (Architecture sections)

📁 2. DATA STORES (RDBMS + NoSQL)

SL-UDI uses hybrid data storage.

Objective

Handle:

- Structured transactional data
 - Massive biometric and logs data
 - High-speed queries
-

Detailed Scope:

Relational Database (RDBMS)

Used for:

- Citizen demographic records
- Enrolment transactions
- Authentication logs
- CRM data

Features required:

- ACID compliance
 - High availability
 - Replication
 - Backup
-

NoSQL Database

Used for:

- Large-scale biometric metadata
- High-volume logs
- Scalable unstructured data

Features:

- Horizontal scaling
- Fast read/write

- Fault tolerance
-

MSI Responsibilities:

- ✓ Schema design
 - ✓ Data partitioning/sharding
 - ✓ Replication setup
 - ✓ Performance tuning
 - ✓ Backup strategies
-

3. INTEGRATION LAYER (MOST IMPORTANT TECHNICALLY)

This connects everything to everything.

Main Components:

-  a) API Gateway (Internal + External)
-

Purpose:

Single controlled entry point for:

- Portals
 - Mobile apps
 - Partner agencies
 - TSPs
-

Functions:

- Request routing
- Authentication & authorization
- Throttling (rate limits)

- Load balancing
 - API versioning
 - Logging
-

MSI Responsibilities:

- ✓ Deploy API gateway platform
 - ✓ Secure all APIs
 - ✓ Monitor usage
-

b) Integration Middleware / ESB

Purpose:

Connect:

MOSIP ↔ Bespoke Apps ↔ ABIS ↔ BI ↔ External systems

Functions:

- Message transformation
 - Routing
 - Orchestration
 - Error handling
-

MSI Responsibilities:

- ✓ Implement ESB/middleware
 - ✓ Ensure reliability
 - ✓ Performance tuning
-

c) Messaging Layer

Purpose:

Handle asynchronous communication such as:

- Enrolment packet flows
 - Notifications
 - Long-running workflows
-

Functions:

- Queues
 - Topics
 - Event-driven processing
-

MSI Responsibilities:

- ✓ Deploy message brokers
 - ✓ Configure durability
 - ✓ Monitor throughput
-

d) ETL / ELT Pipelines

Purpose:

Move data from:

- MOSIP
- CRM
- ABIS
- Portals

→ Into BI & analytics warehouse

Functions:

- Data extraction
 - Cleaning
 - Transformation
 - Loading
-

MSI Responsibilities:

- ✓ Design pipelines
 - ✓ Automate jobs
 - ✓ Handle failures
-

e) Distributed Caching Layer

Purpose:

- Improve performance
- Reduce database load

Used for:

- Frequently accessed identity data
 - Session data
-

MSI Responsibilities:

- ✓ Deploy in-memory cache systems
 - ✓ Configure eviction policies
-

4. SECURITY ARCHITECTURE (VERY HEAVY IN RFP)

Security is **built-in**, not added later.

Core Security Principles:

- Security by design
 - Zero trust
 - Least privilege
 - Encryption everywhere
-

Detailed Security Controls:

a) Encryption

- Data at rest encrypted
 - Data in transit encrypted
 - PKI infrastructure
-

b) Identity & Access Management

- SSO
 - Role-based access
 - MFA for admins
-

c) Network Security

- Firewalls
 - Intrusion detection/prevention
 - Segmentation
-

d) Audit Logging

- Every action logged
 - Tamper-proof logs
 - Centralized monitoring
-



e) Vulnerability Management

- Regular vulnerability scans
 - Penetration testing before go-live
 - Closure of issues
-



f) Risk Management

- Periodic risk assessment
 - Risk register
 - Mitigation plans
-



g) Data Leakage Prevention (DLP) & Endpoint Security

- Monitor 4000+ servers/endpoints
 - Incident response
-



MSI Responsibilities:

- ✓ Implement full security framework
 - ✓ Continuous monitoring
 - ✓ Compliance reporting
 - ✓ Incident response
-



5. OPERATIONS, SUPPORT & SLA MANAGEMENT

SL-UDI is not just built — it must be run continuously.



Support Model:



Levels:

- L1 – Contact Center / basic support
- L2 – Technical support

- L3 – Advanced/OEM support
-



Responsibility split:

Year 1:

MSI/MSP heavily involved

Year 2 & 3:

MSP takes MOSIP core support



Contact Center Setup:

MSI provides:

- CRM software
- IT infrastructure
- Training

GoSL provides:

- Space
 - Manpower
 - Toll-free number
-



Monitoring:

- Application health
 - Infrastructure performance
 - Security alerts
 - SLA metrics
-



MSI Responsibilities:

- ✓ 24x7 monitoring
- ✓ Incident management

- ✓ Patch management
 - ✓ Reporting
-



6. GOVERNANCE & PROJECT MANAGEMENT

Ensures project stays:

- ✓ On schedule
 - ✓ On budget
 - ✓ Compliant
-



Governance Structure:



Project Governance Steering Committee (PGSC)

- Policy direction
 - Risk resolution
 - High-level oversight
-



Project Management Committee (PMC)

- Day-to-day monitoring
 - Reviews
 - Deliverable tracking
-



Technical Working Groups (TWG)

- Domain experts
 - Architecture
 - Legal/privacy
-



MSI PMO Responsibilities:

- End-to-end project plan
- Tools & reporting
- Change control
- Escalation matrix
- Risk management
- Documentation

 SLA Monitoring:

- Technical SLAs
- Project SLAs
- Operations SLAs

 BIG PICTURE OF D)

Layer	Purpose
DC/DR	High availability & continuity
RDBMS/NoSQL	Data backbone
API Gateway	Secure access
Middleware	System integration
Messaging	Async workflows
ETL	Analytics data flow
Caching	Performance
Security	Protection & compliance
Operations	24x7 reliability
Governance	Control & accountability

✓ YOU NOW HAVE FULL SCOPE:

- ✓ A) Bespoke Apps (deep)
- ✓ B) Biometrics & ABIS (deep)
- ✓ C) Process Modules (deep)
- ✓ D) Infrastructure & Security (deep)

This together is the **complete SL-UDI Scope of Work as per RFP**.

If you'd like, I can now:

- 📄 Convert all this into a structured SOW document format
- 📊 Create a simplified end-to-end architecture explanation
- 💬 Prepare interview/exam style Q&A
- 📋 Make comparison MOSIP vs Non-MOSIP responsibilities

Just tell me 👍



Create a simplified end-to-end architecture explanation
and Make comparison MOSIP vs Non-MOSIP
responsibilities

Rakshitha — I'll now do two things clearly and simply (no heavy RFP language):

- ✓ 1) Simplified End-to-End Architecture (how everything connects from citizen → identity → services)
- ✓ 2) Clear MOSIP vs Non-MOSIP responsibility comparison (who does what)

I'll keep it **easy to visualize**, like how a real system works in flow.

🧩 1) SIMPLIFIED END-TO-END SL-UDI ARCHITECTURE (BIG PICTURE)

Think of SL-UDI as **5 connected layers** working together:

LAYER 1: USER INTERFACE LAYER (Where people interact)

This includes:

- Web Portal
- Mobile App
- Enrolment Centers
- Contact Center

What happens here?

Citizens:

- 👉 Pre-register
- 👉 Book appointments
- 👉 Enrol biometrics
- 👉 Track status
- 👉 Use digital services

Agencies:

- 👉 Authenticate users
- 👉 Access services

📡 All requests go through secure APIs

LAYER 2: INTEGRATION & ACCESS LAYER (Traffic controller)

This includes:

- API Gateway
- Middleware (ESB)
- Messaging queues

What happens here?

- ✓ Routes all requests
- ✓ Secures APIs
- ✓ Handles load
- ✓ Connects systems

Example:

Portal → API Gateway → MOSIP

ABIS → Middleware → IDMS

BI → ETL → Data warehouse



LAYER 3: CORE IDENTITY ENGINE (MOSIP)

This is the **brain of the system**.

Includes:

- Pre-Enrolment system
- Enrolment software
- IDMS (workflow engine)
- Unique ID Generator
- Authentication services
- Partner & device management

What happens here?

- ✓ Stores citizen identity
- ✓ Verifies data
- ✓ Runs lifecycle workflows
- ✓ Generates UDI
- ✓ Handles authentication



LAYER 4: BIOMETRIC TRUST ENGINE

Includes:

- Biometric SDK
- BioSDK service
- ABIS
- Manual adjudication

What happens here?

- ✓ Quality check biometrics
- ✓ Deduplication (no duplicates)
- ✓ Matching & verification

IDMS asks ABIS:

👉 "Does this person already exist?"

ABIS answers:

- ✓ Unique
- ✗ Duplicate
- ⚠ Needs human review



LAYER 5: DATA + ANALYTICS + SECURITY

Includes:

- RDBMS + NoSQL databases
- BI & analytics system
- DMS
- Fraud system
- Security controls

What happens here?

- ✓ Stores all data securely
 - ✓ Generates reports
 - ✓ Detects fraud
 - ✓ Manages documents
 - ✓ Enforces compliance
-

SIMPLE END-TO-END FLOW (IN REAL LIFE)

Let's walk through one citizen:

- 1 Citizen pre-registers on portal/mobile**

- 2 Goes to enrolment center → biometrics captured**

- 3 Enrolment packet sent to MOSIP IDMS**

- 4 IDMS sends biometrics to ABIS**

- 5 ABIS checks duplicates**

- 6 If unique → UDI generated**

- 7 UDI activated**

- 8 Citizen can now authenticate for services**

Meanwhile:

- ✓ BI monitors everything
 - ✓ Fraud system checks risks
 - ✓ Security protects all data
-



IN ONE LINE:

Frontend Apps → API Gateway → MOSIP Core → ABIS → Databases + BI + Security



2) MOSIP vs NON-MOSIP RESPONSIBILITIES (VERY IMPORTANT)

This is what many people get confused about.

Let's make it crystal clear.



MOSIP RESPONSIBILITIES (CORE ID PLATFORM)

MOSIP handles identity creation and management

MOSIP DOES:

- ✓ Pre-enrolment system
- ✓ Enrolment client software
- ✓ Identity workflow (IDMS)
- ✓ Unique ID generation
- ✓ Authentication services
- ✓ Partner onboarding
- ✓ Device registration
- ✓ Admin consoles
- ✓ Core security principles

In simple words:



MOSIP = "Digital ID engine"

It ensures:

- One person = one ID
 - Secure identity lifecycle
 - Standardized global platform
-

NON-MOSIP RESPONSIBILITIES (COUNTRY-SPECIFIC ECOSYSTEM)

These are built/customized by MSI.

NON-MOSIP DOES:

- ✓ Web portal
- ✓ Mobile app
- ✓ CRM/helpdesk
- ✓ BI & analytics
- ✓ Document management
- ✓ Billing system
- ✓ Knowledge & training systems
- ✓ Fraud management
- ✓ Process workflows (renewal, lost card etc.)
- ✓ Integration platforms
- ✓ Infrastructure
- ✓ Security operations

In simple words:

👉 Non-MOSIP = "Service ecosystem around the ID"

It ensures:

- Citizen experience
 - Operations
 - Monitoring
 - Revenue
 - Governance
-



SIMPLE COMPARISON TABLE

Area	MOSIP	Non-MOSIP
Pre-registration	✓ Core engine	✗ (only UI uses it)
Enrolment software	✓	✗
Biometric workflow	Partial	✓ ABIS + SDK
Identity lifecycle	✓	✗
Unique ID generation	✓	✗
Authentication engine	✓	✗
Web portal	✗	✓
Mobile app	✗	✓
CRM	✗	✓
BI & analytics	✗	✓
Document system	✗	✓
Fraud system	✗	✓
Infrastructure	✗	✓
Security ops	✗	✓
Process workflows	Partial	✓



VERY EASY WAY TO REMEMBER:



MOSIP = Core Identity Platform



Non-MOSIP = Everything that makes it usable in real life

OR

- 👉 MOSIP builds the ID
- 👉 Non-MOSIP builds the **digital services around the ID**

FINAL ARCHITECTURE LOGIC (ONE GRAPH IN WORDS)

markdown

 Copy code

```
Citizens & Agencies
    ↓
Portal / Mobile / Enrolment
    ↓
API Gateway & Middleware
    ↓
MOSIP Core ID Engine
    ↓
ABIS Biometric Deduplication
    ↓
Databases + BI + Fraud + Security
```

If you want next, I can:

- ✓ Draw this as a clean ASCII architecture diagram
- ✓ Create a presentation-ready explanation
- ✓ Convert into study notes
- ✓ Create interview Q&A

 ChatGPT 5.2 ▾

 ...

- 👉 "Make a diagram"
- 👉 "Make summary notes"
- 👉 "Prepare interview questions"

     ...

